University of British Columbia, Vancouver
Department of Computer Science

CPSC 304 Project Cover Page

Milestone #: 1

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Group Number: 113

Name	Student Number	CS Alias (Userid)	Preferred E-mail Address
Joseph Song	38031209	q1u8s	jsong1618@gmail.com
Ali Javed	24599169	v3w0b	alijavedansari@gmail.com
Runju Huang	38225363	t9g1z	runjuuuuu@gmail.com

By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above.

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

a. What is the domain of the application? Describe it.

The domain of our application is smart kitchen and recipe sharing. The app focuses on helping users manage their household food inventory, plan meals effectively, reduce food waste and share cooking experiences with others.

b. What aspects of the domain are modeled by the database?

The database models three key aspects of this domain: user inventory, recipe and social sharing.

For user inventory, the database stores the ingredients a user currently has at home (this includes quantity, added date and expiration date). Ingredients are also linked to nutritional information which allows the system to calculate nutrition based on items the user adds to their meals.

For recipes, the database models how ingredients are combined to create meals. Each recipe is associated with specific ingredients and quantities which allows the application to estimate total nutritional value and provide the approximate cost per meal.

For social sharing, the database models the exchange of cooking experiences (user posts related to recipes). A post links to a recipe and includes content like reviews and/or cooking tips. This allows other users to view shared posts and easily access the underlying recipe to recreate the meals themselves.

What functionality will the database provide?

The database helps keep track of user accounts (differentiating between trial and paid users) and supports inventory management by storing ingredients, their quantities and expiration dates. The database also allows users to create, store and annotate recipes, link recipes to ingredients and supports meal planning based on the ingredients currently available in their inventory. Additionally, the database supports social features such as the creation of posts and allows for social engagement through reactions. Another functionality supported by the database is receipt uploads with OCR extraction, which allows grocery receipt data to be used for automatically updating a user’s inventory.

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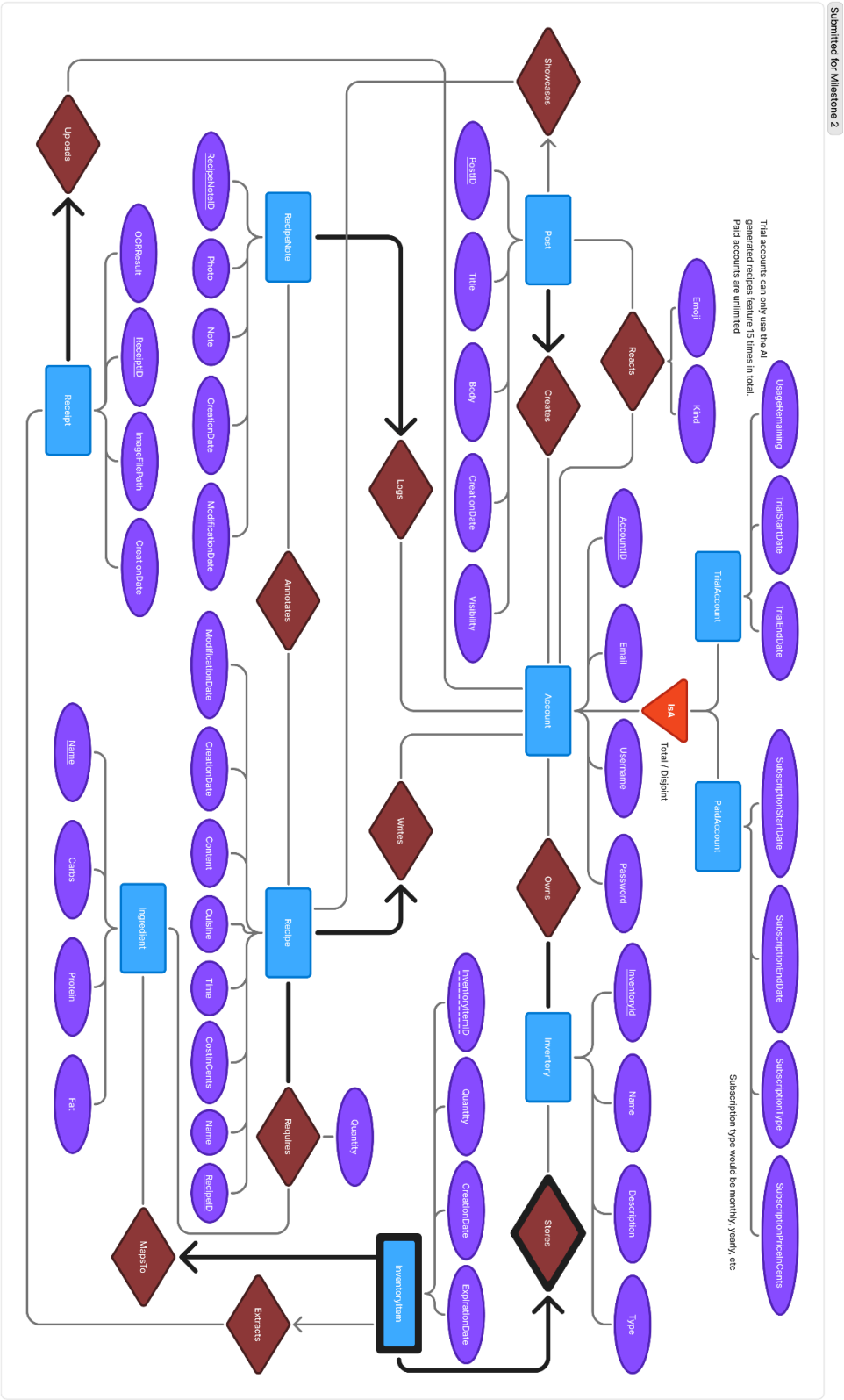
Project Idea

We want to build an app that can track what we have at home and use AI to generate recipes based on it. Since eating out is pretty expensive these days, and we don't have a lot of cooking experience, it would be really helpful to get a recipe based on our available ingredients and then start cooking. We've considered that AI might sometimes generate bad recipes, so we think it would be also useful if users could share recipes they've tried and found to be good. Since the goal is to help us cook at home, we want the inventory to be shared across accounts as well. Because the recipe is based on what we've got at home, we're also considering using AI to extract the ingredient information from the receipt, so we can just take a picture of it and let the AI pre-fill the ingredient list with what they bought.

For the parts involving AI, we'll just call external APIs (e.g., https://ai.google.dev/gemini-api/docs#javascript) and pass a prompt. This would make the AI part trivial and serve primarily as a proof of concept.

Use of AI tools

We didn't use any AI for this milestone

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